

RAJU SANGA

ASIC Verification Engineer | SystemVerilog | UVM | Functional & Coverage-Driven Verification | SVA | ASIC/SoC

Hyderabad, India | +91-8341867409 | sangaraju2003@gmail.com | linkedin.com/in/rajusanga0

PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate and Design & Verification Trainee with hands-on experience building UVM-based, coverage-driven verification environments for digital and mixed-signal ASIC/SoC IP. Verified I2C, SPI, and AMBA/APB protocol blocks to over 95% FSM coverage using constrained-random and directed test strategies, with SystemVerilog Assertions (SVA) for protocol compliance and timing-sensitive checks. Built and verified a parameterized asynchronous FIFO (Gray-code CDC synchronization, 100% functional coverage) and a pipelined 32-bit ALU, using Synopsys VCS for simulation and Verdi for waveform debug, with exposure to Cadence verification tools. Comfortable with CDC/RDC analysis and STA fundamentals across the RTL-to-GDSII flow, and automate regression/log-analysis workflows in Python and shell scripting. Seeking an ASIC Verification Engineer role at Broadcom contributing to pre-silicon functional verification, testbench development, and coverage closure on high-complexity SoC/IP projects.

CORE COMPETENCIES

ASIC/SoC Functional Verification • UVM Testbench Development • Constrained-Random & Coverage-Driven Verification • SystemVerilog Assertions (SVA) • CDC/RDC Verification • Protocol Verification (I2C, SPI, AMBA/APB) • Regression & Debug Automation

TECHNICAL SKILLS

Verification Languages & Methodology: SystemVerilog, UVM (Testbench Architecture, Sequences, Scoreboards, Drivers/Monitors), SystemVerilog Assertions (SVA), Constrained-Random & Coverage-Driven Verification, Assertion-Based Verification, Directed & Regression Testing, Functional Coverage Closure

RTL / Digital & Mixed-Signal Design Knowledge: Parameterized RTL, FSM Design, Synchronous & Asynchronous FIFO, Multi-Clock Domain Design, Pipelined Datapaths, ALU Design, IP Design & Integration

Programming / Scripting: iC, C++, TCL, Perl (working knowledge), Shell Scripting (Linux/Unix)

Protocols Verified: I2C, SPI, AMBA/APB, AXI (Basics)

ASIC / VLSI Flow: ASIC Design Flow, RTL-to-GDSII Overview, CDC/RDC Analysis, Lint Analysis, Synthesis, Static Timing Analysis (STA) Fundamentals, Memory Interfaces, Design Reviews & Documentation

EDA Tools & Platforms: Synopsys VCS (Simulation), Verdi (Waveform Debug), Cadence Verification Suite, Linux/Unix, Git

Soft Skills: Problem Solving, Root-Cause Debugging, Design Documentation, Cross-Functional Collaboration, Individual Contributor & Team Player

PROFESSIONAL EXPERIENCE

Design & Verification Trainee — SumedhaIT, Hyderabad

Jul 2025 – Mar 2026

- Built SystemVerilog/UVM verification environments (drivers, monitors, scoreboards, sequences) for block-level IP validation, covering directed, constrained-random, and coverage-driven test strategies for pre-silicon sign-off.
- Verified I2C and SPI protocol blocks achieving over 95% FSM coverage using Synopsys VCS/Verdi, with exposure to Cadence EDA tools; authored SVA assertions for protocol compliance and timing-sensitive behaviors.
- Debugged complex RTL and protocol-level issues via simulation log analysis and Verdi waveform tracing, applying CDC/RDC analysis, STA fundamentals, AMBA/APB, synthesis, and memory-interface basics in design reviews.
- Used Python and shell scripting to automate regression checks and log parsing, improving verification turnaround and development methodology/quality flow.

DESIGN PROJECTS

Asynchronous FIFO — UVM-Based Functional Verification

SystemVerilog | UVM | Synopsys VCS | Verdi | Cadence | CDC

- Verified a parameterized asynchronous FIFO with Gray-code pointer synchronization and configurable data width/depth for cross-clock-domain data transfer.
- Built a coverage-driven UVM testbench (constrained-random + directed) verifying overflow, underflow, reset, simultaneous read/write, and CDC corner cases; implemented SVA assertions for data integrity and flag validation.
- Achieved 100% functional coverage; resolved CDC corner cases through systematic waveform analysis in Verdi.

32-Bit Pipelined ALU — Functional & Assertion-Based Verification

Verilog | SystemVerilog | Synopsys VCS | Cadence

- Verified a parameterized 32-bit pipelined ALU (arithmetic/logical operations, Carry, Zero, Overflow, Sign, Parity flags) against a directed and corner-case test plan.
- Developed a verification testbench covering overflow/underflow, boundary conditions, and pipeline-stage transitions; validated via simulation, SVA assertions, and corner-case debugging.

EDUCATION

B.Tech, Electronics & Communication Engineering

Graduated: May 2025

Vaageswari College of Engineering, JNTU Hyderabad | CGPA: 6.84 / 10.0